



# UMESH

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## EDUCATION

|                                |           |                                         |      |
|--------------------------------|-----------|-----------------------------------------|------|
| B.TECH(MECHANICAL ENGINEERING) | 2023-2027 | NSUT, New Delhi                         | 6.98 |
| CBSE(Class XII)                | 2022      | Sarvodaya Boys S S S Matiala, New Delhi | 84%  |
| CBSE(Class X)                  | 2020      | Sarvodaya Boys S S S Matiala, New Delhi | 82%  |

## SUMMARY

Mechanical Engineering undergraduate with hands-on experience in automotive CAD/CAE using Siemens NX, Fusion 360, and ANSYS. Experienced in 3D mechanical design, parametric assemblies, and physics-driven engineering simulations, bridging the gap between digital modelling and real-world kinematic and structural behaviour

## PROJECTS

### Thermo-Mechanical Analysis of Disc Brake Rotor | ANSYS Workbench, Siemens NX

- Performed transient thermal and structural FEA of a ventilated, cross-drilled disc rotor under representative braking conditions, evaluating temperature distribution, thermal deformation and von Mises stress.
- Conducting fatigue-life assessment to identify critical stress regions and evaluate rotor durability under cyclic thermo-mechanical loading.

### Double Wishbone Suspension Assembly | Siemens NX

- Designed and assembled a 3D double-wishbone suspension system in Siemens NX, including control arms, steering knuckle, disc brake, wheel and associated mounting features.
- Developed components with consideration for suspension geometry, assembly interfaces, clearances and automotive component integration.

## ACADEMICS & ACHIEVEMENTS

- Secured a position among the **Top 5** scorers in the Class X CBSE Board Examinations at the school level.
- Interested in vehicle dynamics and suspension geometry.

## POSITION OF RESPONSIBILITY

### ASME , NSUT — Recruitment Team Member

Supported the student recruitment process through candidate interviews, applicant communication, and systematic organization, analysis, and shortlisting of candidate data using Excel.

## TECHNICAL SKILLS

- CAD** : Siemens NX(Solid Modelling , Assembly), Fusion 360(T-Spline & Freeform Surface Modelling).
- CAE** : ANSYS Workbench (Fluent, CFX, Structural, Thermal, Rigid dynamics)
- Programming & Scripting** : Python(Automated FEA Workflows via PyAnsys).

## CERTIFICATION/TRAINING

### CAE Certification : Industrial CAE Training program, NSUT

18<sup>th</sup> may – 26<sup>th</sup> June 2026

- Completed an intensive professional program covering advanced CAD modelling, FEA fundamentals, and fluid dynamics using Fusion 360 and ANSYS.
- Engaged in technical mastery workshops led by industry subject matter experts from Autodesk, Curiosity3D, and ARK Infosolutions.